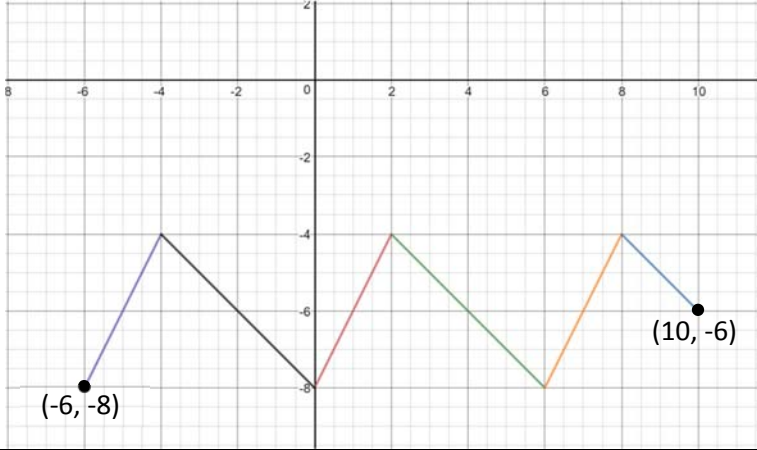
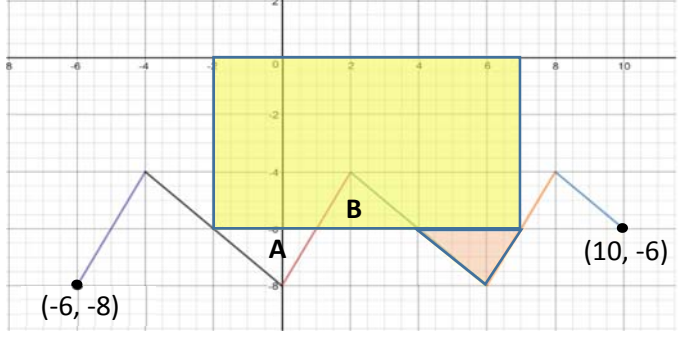
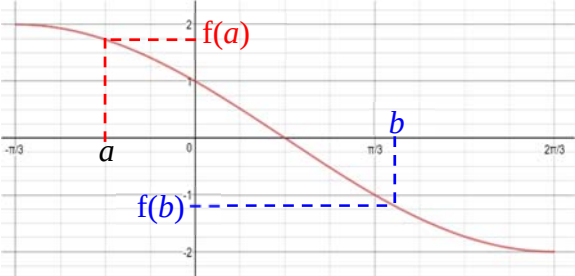


2018 IJC H2 Math Prelim Paper 2 Solution

Qn	Solution
1(i)	$f(-21) + f(49) = f(-21 + 24) + f(49 - 48)$ $= f(3) + f(1)$ $= (-3 - 2) + (2 - 8)$ $= -11$
1(ii)	
1(iii)	Method 1 $\int_{-2}^7 f(x) dx$ $= -\{\text{sum of areas of 4 trapezia}\}$ $= -\left[\frac{1}{2}(6+8)2\right] - \left[\frac{1}{2}(8+4)2\right] - \left[\frac{1}{2}(4+8)4\right] - \left[\frac{1}{2}(8+6)2\right]$ $= -14 - 12 - 24 - 7$ $= -57$ Method 2 $\int_{-2}^7 f(x) dx$ $= \int_{-2}^4 (-x-2) dx + \int_0^2 (2x-8) dx + \int_2^6 (-x-2) dx + \int_6^7 (2x-8) dx$ $= -57$ Method 3 Since areas of triangles A and B are identical, $\int_{-2}^7 f(x) dx = -[\text{Area of rectangle} + \text{area of triangle}]$ $= -\left[9 \times 6 + \frac{1}{2} \times 2 \times 3\right]$ $= -57$ Note: Areas of triangles A and B are identical. 

2(i)	$R = \sqrt{1+3} = 2$ $\alpha = \tan^{-1} \sqrt{3} = \frac{\pi}{3}$ $\cos x - \sqrt{3} \sin x = 2 \cos \left(x + \frac{\pi}{3} \right)$ $R_f = [-2, 2]$
2(ii)	$f : x \mapsto \cos x - \sqrt{3} \sin x, \quad -\frac{\pi}{3} \leq x \leq \frac{2\pi}{3}$ Method 1: $f(x) = 2 \cos \left(x + \frac{\pi}{3} \right) \Rightarrow f'(x) = -2 \sin \left(x + \frac{\pi}{3} \right)$ $-\frac{\pi}{3} \leq x \leq \frac{2\pi}{3} \Rightarrow 0 \leq x + \frac{\pi}{3} \leq \pi$ For $0 < x + \frac{\pi}{3} < \pi \Rightarrow \sin \left(x + \frac{\pi}{3} \right) > 0$ $f'(x) < 0$ When $x + \frac{\pi}{3} = 0$ or π, $\sin \left(x + \frac{\pi}{3} \right) = 0$ $\Rightarrow f'(x) = 0$ The end-points are stationary points. Thus $f(x)$ decreases as x increases. Method 2: Sketch the graph of $y = f(x)$.  From the graph, for $-\frac{\pi}{3} \leq a < b \leq \frac{2\pi}{3}$, $f(a) > f(b).$ Thus $f(x)$ decreases as x increases Method 3: For $-\frac{\pi}{3} \leq a < b \leq \frac{2\pi}{3}$ $f(a) - f(b) = 2 \cos \left(a + \frac{\pi}{3} \right) - 2 \cos \left(b + \frac{\pi}{3} \right)$ $= -4 \sin \left[\frac{1}{2} \left(a + b + \frac{2\pi}{3} \right) \right] \sin \left[\frac{1}{2} (a - b) \right]$ $= -4 \sin \left(\frac{a+b}{2} + \frac{\pi}{3} \right) \sin \left[\frac{1}{2} (a - b) \right]$ Since $-\frac{\pi}{3} < \frac{a+b}{2} < \frac{2\pi}{3}$, $\Rightarrow 0 < \frac{a+b}{2} + \frac{\pi}{3} < \pi$ $\sin \left(\frac{a+b}{2} + \frac{\pi}{3} \right) > 0$

	Since $-\frac{\pi}{3} \leq a < b \leq \frac{2\pi}{3}$, $\Rightarrow -\frac{\pi}{3} - \frac{2\pi}{3} \leq a - b < 0$ $-\frac{\pi}{2} \leq \frac{a-b}{2} < 0$ $-1 \leq \sin\left(\frac{a-b}{2}\right) < 0$ Thus $f(a) - f(b) > 0$ i.e. For $a < b$, $f(a) > f(b)$ Therefore, $f(x)$ decreases as x increases.
2(iii)	Let $y = 2 \cos\left(x + \frac{\pi}{3}\right)$ $\cos\left(x + \frac{\pi}{3}\right) = \frac{y}{2}$ $x = \cos^{-1}\left(\frac{y}{2}\right) - \frac{\pi}{3}$ $f^{-1}(x) = \cos^{-1}\left(\frac{x}{2}\right) - \frac{\pi}{3}, \quad -2 \leq x \leq 2$
2(iv)	$fg(x) = x - 3, \quad 1 \leq x \leq 3$ $f^{-1}fg(x) = f^{-1}(x - 3)$ $g(x) = \cos^{-1}\left(\frac{x-3}{2}\right) - \frac{\pi}{3}$ <u>Alternatively</u> $f(g(x)) = x - 3, \quad 1 \leq x \leq 3$ $2 \cos\left(g(x) + \frac{\pi}{3}\right) = x - 3$ $g(x) + \frac{\pi}{3} = \cos^{-1}\left(\frac{x-3}{2}\right)$ $g(x) = \cos^{-1}\left(\frac{x-3}{2}\right) - \frac{\pi}{3}$
3(i)	$\frac{dV}{dt} = 1200 - A\sqrt{h}$, where A is a positive constant. $V = \pi r^2 h = 3000h \Rightarrow \frac{dV}{dt} = 3000 \frac{dh}{dt}$ $\frac{dh}{dt} = \frac{1}{3000}(1200 - A\sqrt{h})$ $= \frac{1200}{3000} - \frac{A}{3000}\sqrt{h}$ $= 0.4 - k\sqrt{h} \quad \text{where } k = \frac{A}{3000} > 0$

3(ii)	When $h = 36$, $A\sqrt{36} = 360 \Rightarrow A = 60$ $k = \frac{60}{3000} = 0.02$
3(iii)	$h = (20 - x)^2$ L L (1) $\Rightarrow \frac{dh}{dt} = -2(20 - x) \frac{dx}{dt}$ L L (2) Substituting (1) & (2) into $\frac{dh}{dt} = 0.4 - 0.02\sqrt{h}$, $-2(20 - x) \frac{dx}{dt} = 0.4 - 0.02(20 - x)$ $(x - 20) \frac{dx}{dt} = 0.01x$ L L (*) $\int \frac{x-20}{x} dx = \int 0.01 dt$ $\int 1 - \frac{20}{x} dx = \int 0.01 dt$ $x - 20 \ln x = 0.01t + C$ $t = 100[x - 20 \ln x - C]$ $= 100[(20 - \sqrt{h}) - 20 \ln 20 - \sqrt{h} - C]$ $= 100[-\sqrt{h} - 20 \ln 20 - \sqrt{h} + C']$ where $C' = 20 - C$ When $t = 0, h = 0 \Rightarrow C' = 20 \ln 20$ $\frac{dh}{dt} > 0 \Rightarrow 0.4 - 0.02\sqrt{h} > 0$ $0.02(20 - \sqrt{h}) > 0$ $20 - \sqrt{h} > 0$ Thus $t = 100 \left[-\sqrt{h} + 20 \ln \frac{20}{20 - \sqrt{h}} \right]$
3(iv)	When $h = 100$, $t = 100[-10 + 20 \ln 2] = 386.294$ It takes 6 mins and 26 seconds for the height to reach 100 cm.
4	Method 1: $g(x) = af(0.5x - 1)$ Sequence of transformations involved: A: A scaling of the graph of $y = f(x)$ parallel to the y-axis with scale factor a B: A translation of 1 unit in the positive x-direction C: A scaling parallel to the x-axis with scale factor 2. Method 2: $g(x) = af(0.5(x - 2))$ Sequence of transformations involved: A: A scaling of the graph of $y = f(x)$ parallel to the y-axis with scale factor a B: A scaling parallel to the x-axis with scale factor 2. C: A translation of 2 units in the positive x-direction Acceptable alternative order of sequence: BCA.

4(i)	Volume $= \pi \int_0^2 \frac{a^2}{1+(0.5x-1)^2} - \frac{1}{1+x^2} dx$ $= \pi \left[\frac{a^2}{0.5} \tan^{-1}(0.5x-1) - \tan^{-1} x \right]_0^2$ $= \pi \left[\left(2a^2 \tan^{-1} 0 - \tan^{-1} 2 \right) - \left(2a^2 \tan^{-1}(-1) - \tan^{-1} 0 \right) \right]$ $= \pi \left(\frac{\pi}{2} a^2 - \tan^{-1} 2 \right)$
4(ii)	$y = \frac{1}{\sqrt{1+x^2}} \Rightarrow x^2 = \frac{1}{y^2} - 1$ Volume of 2nd sculpture $= \text{Vol of cylinder} + \pi \int_{\frac{1}{\sqrt{5}}}^1 \frac{1}{y^2} - 1 dy$ $= \pi \left(2^2 \right) \frac{1}{\sqrt{5}} + \pi \left[-\frac{1}{y} - y \right]_{\frac{1}{\sqrt{5}}}^1$ $= \pi \left(\frac{4}{\sqrt{5}} \right) + \pi \left[(-2) - \left(-\sqrt{5} - \frac{1}{\sqrt{5}} \right) \right]$ $= 2\pi(\sqrt{5} - 1)$
4(iii)	$\pi \left(\frac{\pi}{2} a^2 - \tan^{-1} 2 \right) \geq 50 \left[2\pi(\sqrt{5} - 1) \right]$ $a^2 \geq \frac{2}{\pi} \left[100(\sqrt{5} - 1) + \tan^{-1} 2 \right]$ $a \geq 8.91 \quad \text{since } a > 2$ Smallest integer value of a is 9
5	$P(X \leq 1) = 0.15$ $\binom{25}{0} (1-p)^{25} + \binom{25}{1} p(1-p)^{24} = 0.15$ $(1-p)^{25} + 25p(1-p)^{24} = 0.15$ $\left[\text{or } (1-p)^{24} (1+24p) = 0.15 \right]$ Using GC, $p = 0.12865$ $= 0.129$ (to 3 s.f.) $\text{Var}(X) = np(1-p)$ $= 25(0.12865)(1-0.12865)$ $= 2.80$ (to 3 s.f.)

6(i)	AA EE I PP R C T c _ c _ c _ c _ _ c _ c _ c _ c Number of ways = $\left(\frac{5!}{2!2!}\right)\left(\frac{5!}{2!}\right) \times 2! = 3600$
6(ii)	E _ _ E _ _ _ _ _ Case 1 : 2 As No of ways = $\frac{7!}{2!} = 2520$ Case 2 : 1A with 1P No of ways = $2!7! = 10080$ Case 3 : 1A without P No of ways = $({}^4C_1 \times 2!) \times \frac{7!}{2!} = 20160$ Total no of ways = 32760
7(i)	Given $\text{Var}(X) = 0.75$, $E(X^2) = (1)^2(3p) + (2)^2(2p) + (3)^2(p) = 20p$ $E(X) = 1(3p) + (2)(2p) + (3)(p) = 10p$ $20p - (10p)^2 = 0.75$ $100p^2 - 20p + 0.75 = 0$ $p = \frac{1}{20}$ or $p = \frac{3}{20}$ (Reject since $0 < p < \frac{1}{10}$) $\therefore p = \frac{1}{20}$ (or 0.05), $E(X) = \frac{1}{2}$ or 0.5
7(ii)	$S = X_1 + X_2 + X_3 + \dots + X_n$ Since n is large, by Central Limit Theorem, $S : N(0.5n, 0.75n)$ approximately. <u>Method 1: Algebraic method</u> $P(S > 150) \geq 0.75$ $P\left(Z < \frac{150 - 0.5n}{\sqrt{0.75n}}\right) \leq 0.25$ $\frac{150 - 0.5n}{\sqrt{0.75n}} \leq -0.6744897$ $150 - 0.5n \leq -0.6744897\sqrt{0.75n}$ $0.5n - 0.6744897\sqrt{0.75n} - 150 \geq 0$ $\sqrt{n} \leq -16.746$ (reject since $\sqrt{n} > 0$) or $\sqrt{n} \geq 17.914$ $n \geq 320.93$ Thus $\{n : n \in \mathbb{C}^+, n \geq 321\}$

Method 2: Using GC (table)

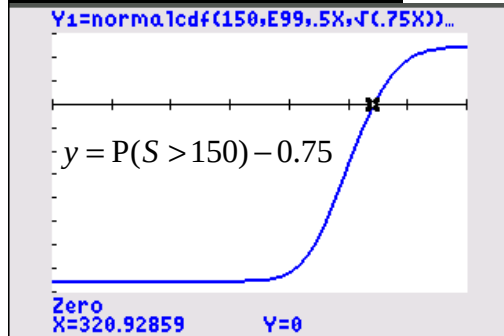
$$P(S > 150) \geq 0.75$$

$$\text{When } n = 320, P(S > 150) = 0.7407 < 0.75$$

$$\text{When } n = 321, P(S > 150) = 0.7507 > 0.75$$

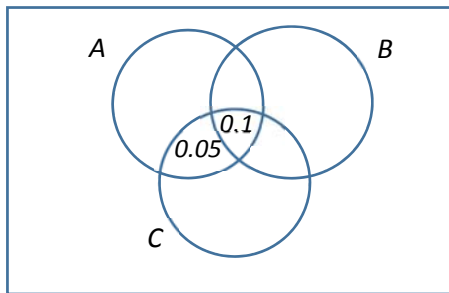
$$\text{When } n = 322, P(S > 150) = 0.7605 > 0.75$$

$$\text{Thus } \{n: n \in \mathbb{C}^+, n \geq 321\}$$

Method 3: Using GC (graph)

From the graph, $n \geq 320.92859$

$$\text{Thus } \{n: n \in \mathbb{C}^+, n \geq 321\}$$

8(i)

Given $P(B|C) = 0.5$,

$$P(B|C) = \frac{P(B \cap C)}{P(C)}$$

$$0.5 = \frac{P(B \cap C)}{0.35}$$

$$P(B \cap C) = 0.175$$

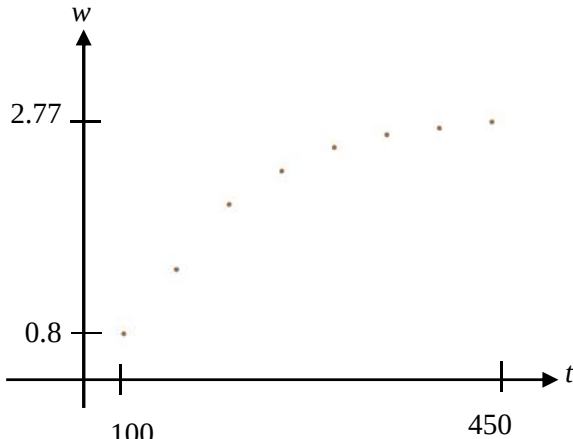
$$\begin{aligned} P(A' \cap B \cap C) &= 0.175 - 0.1 \\ &= 0.075 \end{aligned}$$

8(ii)

Since A and B are independent events,

$$\begin{aligned} P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\ &= P(A) + P(B) - P(A) \times P(B) \\ &= 0.5 + 0.45 - 0.5(0.45) \\ &= 0.725 \end{aligned}$$

8(iii)	$P(A' \cap B' \cap C')$ $= 1 - P(A \cup B \cup C)$ $= 1 - 0.35 - (0.5 - 0.15 - x) - (0.45 - 0.175 - x) - x$ $= 0.025 + x$ Max x occurs when $x + 0.1 + 0.075 = 0.45$ $x = 0.275$, greatest possible value of $P(A' \cap B' \cap C') = 0.3$ when $x = 0$, least possible value of $P(A' \cap B' \cap C') = 0.025$
9(i)	$H_0 : \mu = 250$ $H_1 : \mu < 250$ where μ represents the population mean volume of tomato juice per bottle.
9(ii)	Assume that H_0 is true. Since $n = 50$ is large, by CLT, $\bar{X} \sim N\left(250, \frac{10^2}{50}\right)$ approximately. Using GC, $p\text{-value} = 0.0385498886 \approx 0.0385$ (3 s.f.) There is 0.0385 probability of drawing a random sample of 50 bottles of tomato juice and obtaining a sample mean volume of 247.5 ml or less, assuming that the population mean volume is 250 ml.
9(iii)	No, it is not necessary to assume a normal distribution for the test to be valid, since $n = 50$ is large, Central Limit Theorem can be applied for $\bar{X}$ to be normally distributed.
9(iv)	Unbiased estimate of population mean of smaller bottles of tomato juice $= \frac{10757}{60} = 179.283 \approx 179$ (3 s.f.) Unbiased estimate of population variance $= \frac{1}{59} \left[1931597 - \frac{10757^2}{60} \right]$ $= 51.63022599$ ≈ 51.6 (3 s.f.)
9(iv)	$H_0 : \mu = \mu_0$ $H_1 : \mu \neq \mu_0$ Test at 5% significance level. Assuming H_0 is true, since $n = 60$ is large, by CLT, $\bar{Y} \sim N\left(\mu_0, \frac{51.6302}{60}\right)$ approximately

	Test statistic $Z = \frac{\bar{Y} - \mu_0}{s / \sqrt{60}} \sim N(0, 1)$ H_0 is not rejected $\Rightarrow$ The test statistic lies outside the critical region. $-1.959963986 < \frac{179.283 - \mu_0}{\sqrt{\frac{51.630226}{60}}} < 1.959963986 \quad \text{L L } (*)$ $-1.959963986 \left(\sqrt{\frac{51.630226}{60}} \right) < 179.283 - \mu_0 < 1.959963986 \left(\sqrt{\frac{51.630226}{60}} \right)$ $-1.8181281 < 179.283 - \mu_0 < 1.8181281$ $-1.8181281 - 179.283 < -\mu_0 < 1.8181281 - 179.283$ $177.4652052 < \mu_0 < 181.1014614$ $177.5 < \mu_0 < 181.1 \quad (1 \text{ d.p.})$
10(i)	$r = 0.925$ (3 s.f.) Acceptable answers: - As the pmcc value is close to 1, indicating a strong positive <u>linear</u> correlation, it suggests that a linear model is appropriate. - A linear model with positive linear correlation would suggest that the weight loss may exceed 100%, which is impossible. Thus a linear model is not appropriate.
10(ii)	 (A) $w = a + b \ln t$, from the scatter diagram, as length of time increases, percentage of weight loss also increases at a decreasing rate.
10(iii)	$w = -5.31 + 1.35 \ln t$ The product moment correlation coefficient between $\ln t$ and w is $r = 0.9828402622$ $= 0.983 \quad (3 \text{ s.f.})$ This pmcc value of 0.983 is closer to 1 than the earlier pmcc value of 0.925, indicating stronger positive linear correlation between w and $\ln t$ compared to the linear model.

10(iv)	$2.4 = -5.3074 + 1.3522 \ln t$ $\ln t = \frac{2.4 + 5.3074}{1.3522}$ $t = e^{5.6999}$ $t = 298.84 \approx 299 \text{ (3 s.f.)}$ This estimate is reliable since (1) The estimate is an interpolation, because $w = 2.4$ is within the data range of w. (2) the product moment correlation coefficient between $\ln t$ and w is $r = 0.983$ which is very close to 1, showing a very strong positive linear correlation between $\ln t$ and w.
10(v)	1 day = 24 hours $w = -5.31 + 1.35 \ln(24t)$
11 (a)(i)	$\mu = \frac{40 + 50}{2} = 45$ $P(T < 40) = 0.36$ $P\left(Z < \frac{40 - 45}{\sigma}\right) = 0.36$ $\frac{-5}{\sigma} = -0.3584588$ $\sigma = 13.9486 = 13.9 \text{ (to 3 s.f.)}$
11 (a)(ii)	$P(T \leq 50) = 1 - 0.36 = 0.64$ Let C be the number of houses, out of 18, for which it takes at most 50 min to install an electricity meter. $C \sim B(18, 0.64)$ $P(5 \leq C < 10) = P(C \leq 9) - P(C \leq 4)$ $= 0.160 \text{ (3 s.f.)}$
(b)(i)	Let E be the electricity consumption of the households in Central City $E \sim N(520, 35^2)$ $P(E > 500) = 0.7161454588 = 0.716 \text{ (to 3 s.f.)}$
(b)(ii)	Required probability $= [P(E < 500)]^2$ $= [1 - 0.7161454588]^2$ $= 0.0806 \text{ (to 3 s.f.)}$
(b)(iii)	Part (ii) is a subset of the event where the total electricity consumption for two randomly chosen households is less than 1000 kWh.

(b)(iv)	Let S be the electricity consumption of the households in Star City. $S \sim N(475, 25^2)$ $X = 0.18E \Rightarrow E(X) = 0.18 \times 520 = 93.6$ $\text{Var}(X) = 0.18^2 \times 35^2 = 39.69$ $Y = 0.15S \Rightarrow E(Y) = 0.15 \times 475 = 71.25$ $\text{Var}(Y) = 0.15^2 \times 25^2 = 14.0625$ $E(4Y - 3X) = 4 \times 71.25 - 3 \times 93.6 = 4.2$ $\text{Var}(4Y - 3X) = 4^2 \times 14.0625 + 3^2 \times 39.69$ $= 582.21$ $4Y - 3X \sim N(4.2, 582.21)$ $P(4Y - 3X > 7) = 0.454 \quad (3 \text{ s.f.})$ It means that there is 0.454 probability that 4 times the electricity bill of a randomly chosen household in Star City exceeds 3 times the electricity bill of a randomly chosen household in Central City by more than \$7.
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